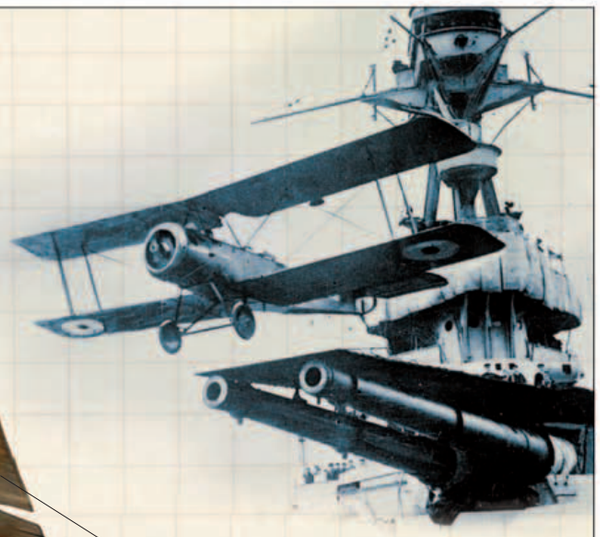
WEAPONS AND TECHNOLOGY

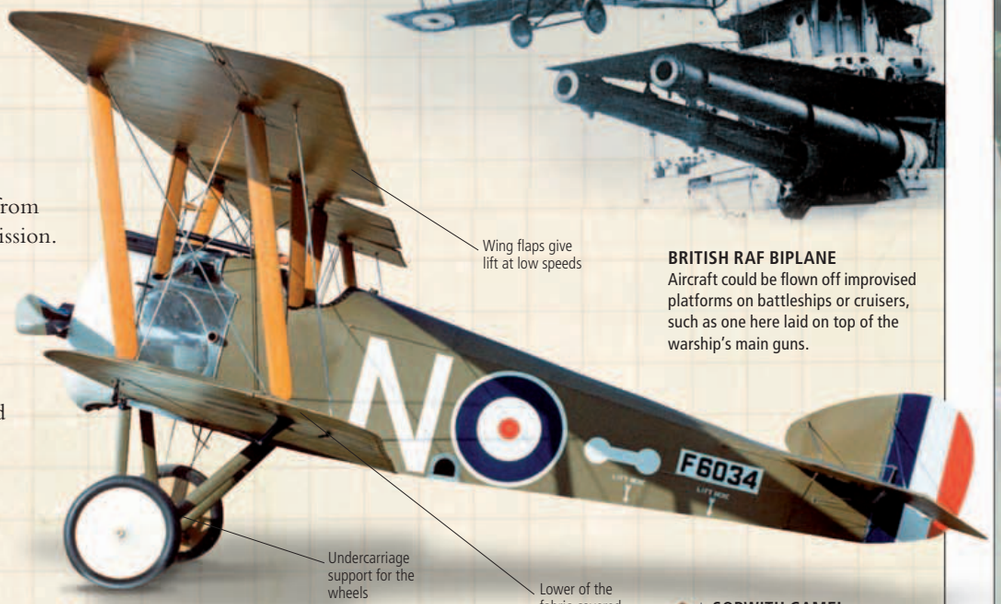
EARLY NAVAL AVIATION

AMERICAN FLIER EUGENE ELY demonstrated in 1910–11 that it was possible to take off from and land on the deck of a ship. Navies were quick to see the usefulness of aircraft, especially for reconnaissance, and adopted both airships and heavier-than-air machines. In 1911 the French navy introduced the first ship to carry float planes and in 1914 Britain's Royal Naval Air Service was founded.



EARLY NAVAL AIRCRAFT

Float planes were used from the start of World War I. Lowered over the side of a seaplane tender to take off from the water, they were lifted back on board after their mission. Unfortunately, the float planes could only operate in optimum weather conditions. A heavy swell made landing and take-off impossible. The lightweight wood-and-canvas aircraft of the period were relatively easy to launch from an improvised platform on a ship, but safe landing was only just being mastered as the war ended. Thus even ship-launched machines had to be retrieved from the sea after a mission.



BRITISH RAF BIPLANE
Aircraft could be flown off improvised platforms on battleships or cruisers, such as one here laid on top of the warship's main guns.

SOPWITH CAMEL
The Sopwith 2F.1 could be split in two just behind the cockpit so it could be more easily moved and stored on the carriers of the period.

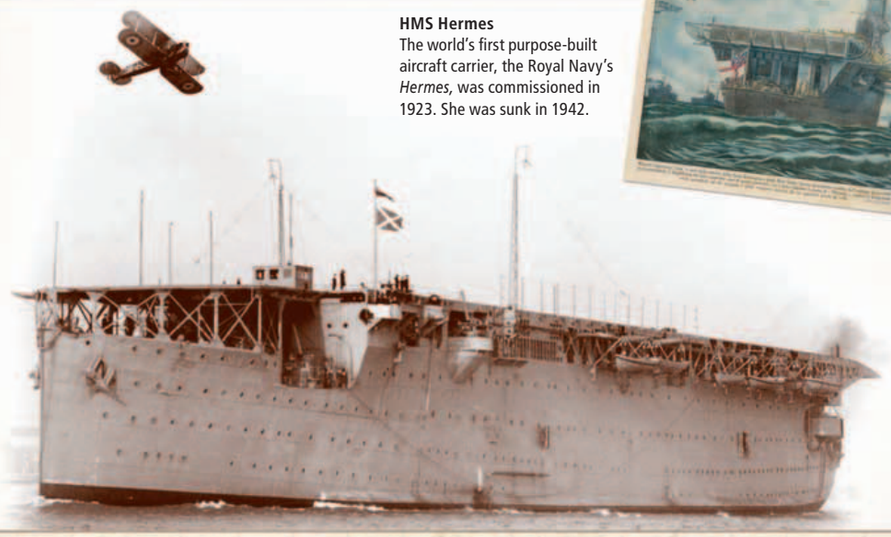


SHORT 184 SEAPLANE
The only plane to participate in the battle of Jutland, the 184 was also the first—on August 12, 1915—to sink a ship with an air-launched torpedo.



CARRIER CONVERSIONS

The first carriers evolved from experiments by the Royal Navy Air Service in 1916. The decks of the battlecruiser *Furious* were cleared to make a platform for take-off and landing, but it was found that the funnel and central superstructure created insuperable problems for pilots. In 1918 the British converted an Italian liner into the carrier *Argus*, with a funnel tucked away at the end of a long unobstructed deck. The US Navy followed this example with the converted collier *Langley*—dubbed the "covered wagon"—in 1922. A superstructure offset to starboard was generally adopted in the 1920s, but most carriers continued to be conversions— the US Navy's *Lexington* and *Saratoga*, for example, started life as battlecruisers.



HMS Hermes
The world's first purpose-built aircraft carrier, the Royal Navy's *Hermes,* was commissioned in 1923. She was sunk in 1942.

HMS Argus
Built in 1916, the British aircraft carrier *Argus* was the first carrrier to have a flush deck that let wheeled planes take off and land safely.